

Element H: Prototype Testing and Data Collection

Requirement.	Testing Set up and Procedure.	Test Cases and Number of Trials.	Expected Range of Results	Pass/Fail Criteria, Goal
Sturdiness test: The book stand must remain intact and stable when holding books of various sizes and weights.	1)Place the book on a flat surface. 2)Gradually place books of increasing weight and size onto the stand. (light, paperback, standard textbook, large hardcover, etc. 3)Observe how the stand responds to each weight level. (if it shifts, sags, collapses 4)Record observations and structural integrity after each trial	Trial 1: Light paper back. (3 times) Trial 2: Standard Textbook (3 times) Trial 3: Large hardcover. (3 times)	Stand may - Remain fully stable - Slightly flex under heavier books - Begin to bend collapse or detach parts under excessive weight.	Fail: The book stand collapses, deforms significantly, or parts detach under normal book weight. Pass: The Stand supports each book type without damage. Goal: The stand holds all types of books without shifting or damage in all 9 trials.
Durability test: The bookstand must be durable and withstand repeated use.	1)Apply downward force at various angles to simulate frequent book placements and adjustments. 2)Repeat pressure applications in the same spots over 10 trials to test durability 3)Monitor for any wear, cracking, or structural weakness overtime.	5 trials simulating daily classroom use.	Stand May: Remain intact - Show signs of wear but no breakage - Crack or weaken at stress points.	Fail: Visible damage, cracking, or failure by trial 5. Pass: Structure remains sound with minimal wear. Goal: No visible wear or weakening across all trials.
Stability Test: The bookstand must be adjustable/moveable without using stability.	1)Adjust the stand to different angles/positions . 2)Place a book on each setting to test ability	10 Trials across 3 Angle settings (Low, Medium, High)	Stand May: -Hold position firmly. -Slightly shift under Pressure - Collapse if angle is setting is weak.	Fail: Stand cannot hold position or collapses in any angle. Pass: Stand adjusts and remains stable in

	3)Observe any slipping instability or inability to maintain the position.			all positions. Goal: Book stand supports book stably in every angle setting with no slippage.
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Feedback From Field Experts.

We consulted with 3 Student field experts who frequently use the school library and classroom tools. Their experience using book stands, heavy textbooks, and digital resources helped us refine our testing plan to better reflect real-world classroom usage and needs.

Field Experts Consulted:

1. Miracle Davis - A Sophomore and avid reader who used the library bookstands to study large textbooks for AP Courses.
 - *Feedback:* Ensure the book stand can hold open, large, heavy books without sliding. Include testing for stability while turning pages.
2. Esongkuo Ayompe - A Senior and Student tutor for NHS who regularly steps up bookstands for tutoring sessions in the library.
 - *Feedback:* Add testing for angle adjustments, since users change angles often when sharing materials with others.
3. Faith Adewole - A Sophomore who frequently reads and uses library stands to pick out books.
 - *Feedback:* Include pressure tests for repeated use and movement, since students often press down or lean on the stand while working.

Revised Testing Details That Incorporate Field Expert Feedback

Requirement (Must be moveable, sturdy, and durable)	Testing setup and procedure.	Test cases and expected number of trials.	Expected range of results.	Pass/Fail Criteria, Goal.
The book stand must be durable with frequent classroom/library use. (Durability)	1) Simulate repeated classroom use by pressing down on the center and edge of the stand 10 times. 2) Use a 5kg force to replicate student pressure when leaning or writing. 3) Observe for any cracks, loosening, or damage.	10 durability trials (force applications)	Stand may: - Remain intact - Show light surface wear - Crack or loosen	Fail: Visible cracks, unstable joints, or part failure. Pass: Fully intact and functional. Goal: No structural damage after repeated use.
The book stand must remain intact and stable when holding books of various sizes and during	1) Place the book stand on a flat desk. 2) Test with books of increasing size/weight	Trial 1: Light book (10 trials) - Trial 2: Standard textbook (10 trials) - Trial 3: Large hardcover	Stand may: - Remain stable - Slightly shift when turning pages - Slide or lean excessively	Fail: Stand slides, collapses, or parts detach. Pass: Stand supports each book without

page-turning. (Sturdiness)		(10 trials)		shifting. Goal: Stand holds books and allows page-turning without movement in all 10 trials.
The book stand must adjust to multiple angles and stay stable during use. (Stability)	Adjust stand to 3 angles: 30°, 60°, and 90°. 2) At each angle, place a textbook and turn 5 pages. 3) Repeat for 10 trials at each angle. 4) Observe if the stand shifts or collapses.	10 trials per angle (30°, 60°, 90°)	Stand may: - Hold steady - Slightly adjust with pressure - Collapse or change angle unexpectedly	Fail: Stand does not hold angle or collapses. Pass: Holds angle and book weight throughout test. Goal: Maintains position and weight